

SOLUTION OF TRIANGLE

JEE EXERCISES

Exercise J-I

BASED ON BASIC LAWS (SINE LAW, COSINE RULE, PROJECTION FORMULA, TANGENT RULE, AREA OF TRIANGLE)

1. In a triangle ABC, right angled at C, $\tan A + \tan B$ is equal to :

त्रिभुज ABC में, C कोण समकोण है तो $\tan A + \tan B$ का मान होगा :

- (A) $a + b$ (B) $\frac{c^2}{ab}$ (C) $\frac{a^2}{bc}$ (D) $\frac{b^2}{ac}$

Ans. B

Sol. $A + B = 90^\circ \Rightarrow \tan A + \frac{1}{\cot B} = \tan A + \frac{1}{\tan A} = \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$

2. In a triangle ABC if $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc}$, then A is :

- (A) an acute angle (B) an obtuse angle (C) a right angle (D) equal to $B - C$

यदि त्रिभुज ABC में $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc}$, तो A होगा :

- (A) एक न्यून कोण (B) एक अधिक कोण (C) एक समकोण (D) $B - C$ के बराबर

Ans. C

Sol. $\Rightarrow \frac{b^2 + c^2 - a^2 + c^2 + a^2 - b^2 + a^2 + b^2 - c^2}{2abc} = \frac{a}{bc} \Rightarrow \frac{a^2 + b^2 + c^2}{2abc} = \frac{a}{bc}$
 $\Rightarrow \angle A = 90^\circ$ (C)

3. If in a ΔABC , $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, then :

- (A) ΔABC may be a scalene triangle (B) ΔABC is a right triangle
(C) ΔABC is an obtuse angled triangle (D) ΔABC is an equilateral triangle

यदि ΔABC में, $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, हो तो :

- (A) ΔABC एक विषमबाहु त्रिभुज हो सकता है (B) ΔABC समकोण त्रिभुज
(C) ΔABC अधिक कोण त्रिभुज (D) ΔABC समबाहु त्रिभुज है

Ans. D

Sol. Use : $x^3 + y^3 + z^3 - 3xyz = (1/2)(x + y + z)[(x - y)^2 + (y - z)^2 + (z - x)^2]$ where $x = \sin A$ etc.

Hint: either $\sin A + \sin B + \sin C = 0$ (which is not possible) or $\sin A = \sin B = \sin C \Rightarrow$ equilateral

4. In a triangle ABC, $a : b : c = 4 : 5 : 6$. Then $3A + B =$

यदि त्रिभुज ABC, $a : b : c = 4 : 5 : 6$ है तो $3A + B =$

- (A) $4C$ (B) 2π (C) $\pi - C$ (D) π

Ans. D

Sol. $a : b : c = 4 : 5 : 6$

$$\Rightarrow \cos A = \frac{25 + 36 - 16}{2 \times 6 \times 5} = \frac{45}{2 \times 6 \times 5} = \frac{3}{4}$$

$$\cos B = \frac{16 + 36 - 25}{2 \times 4 \times 6} = \frac{27}{2 \times 4 \times 6} = \frac{9}{16}$$

$$\cos C = \frac{16 + 25 - 36}{2 \times 4 \times 5} = \frac{5}{2 \times 4 \times 5} = \frac{1}{8}$$

$$\Rightarrow \cos 2A = 2 \cos^2 A - 1 = 2 \times \frac{9}{16} - 1 = \frac{1}{8} = \cos C$$

$$\Rightarrow C = 2A \Rightarrow A + B + C = 3A + B = \pi \quad (D)$$

5. The sides of a triangle are 17, 25 and 28. The length of the largest altitude is :

यदि किसी त्रिभुज की भुजाएँ 17, 25 तथा 28 तो सबसे बड़ा शीर्ष लम्ब होगा :

(A) 15 (B) $84/5$ (C) $420/17$ (D) $210/17$

Ans. C

Sol. $\Rightarrow l_1 = 17 \sin B = 25 \sin C$

$$l_2 = 17 \sin A = 28 \sin C$$

$$l_3 = 25 \sin A = 28 \sin B$$

$$A > B > C$$

$$\Rightarrow l_3 > l_2 > l_1$$

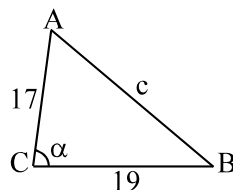
$$\text{Now, } \cos A = \frac{17^2 + 25^2 - 28^2}{2 \times 17 \times 25} = \frac{289 - 3 \times 53}{17 \times 50} = \frac{289 - 159}{17 \times 50} = \frac{13}{85}$$

$$\Rightarrow \sin A = \frac{\sqrt{72 \times 98}}{85} = \frac{6\sqrt{2} \times 7\sqrt{2}}{85} = \frac{84}{85}$$

$$\Rightarrow l_3 = 25 \times \frac{84}{85} = \frac{420}{17} \quad (C)$$

6. Consider the triangle pictured as shown. If $0 < \alpha < \pi/2$ then the number of integral values of c is :

चित्र में एक त्रिभुज दिया गया है तथा यदि $0 < \alpha < \pi/2$ हो तो c के पूर्णांक मानों संख्या होगी :



(A) 35

(B) 23

(C) 24

(D) 25

Ans. B

Sol. $\cos C = \frac{(19)^2 + (17)^2 - c^2}{2 \cdot 19 \cdot 17} > 0$

$$\therefore c^2 < (19)^2 + (17)^2$$

$$c^2 < (18 + 1)^2 + (18 - 1)^2$$

$$c^2 < 2(18^2 + 1)$$

$$c^2 < 650 \quad \dots(1)$$

$$c > 2 \quad \text{and } c < 26$$

$$\therefore 2 < c < 26 \quad \dots(2)$$

$$4 < c^2 < 676 \quad \dots(3)$$

$$\therefore c \text{ is } 3, 4, 5, \dots, 25$$

$$\therefore \text{ number of integral values of } c \text{ is } 23$$

7. Triangle ABC has $BC = 1$ and $AC = 2$. The maximum possible value of the angle A is :

त्रिभुज ABC में $BC = 1$ तथा $AC = 2$ है तो A की सम्भव सबसे बड़ा मान होगा :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$ **Ans.** A

Sol. $\Rightarrow \frac{\sin A}{1} = \frac{\sin B}{2} \Rightarrow \sin A = \frac{\sin B}{2} \Rightarrow \sin A|_{\max} = \frac{1}{2} \Rightarrow A_{\max} = 30^\circ = \frac{\pi}{6} \text{ (A)}$

8. Let ABC be a triangle right angled at C. The value of $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ ($b + c \neq 1, c - b \neq 1$) equals :

माना त्रिभुज ABC एक C पर समकोण त्रिभुज है तो $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ ($b + c \neq 1, c - b \neq 1$) का मान होगा :

(A) 1

(B) 2

(C) 3

(D) $1/2$ **Ans.** B

Sol. $\frac{1}{\log_{c-b} a} + \frac{1}{\log_{b+c} a} = \log_a (c - b) + \log_a (c + b) = \log_a (c^2 - b^2) = \log_a (a^2) = 2 \quad \text{(B)}$

9. In a triangle ABC, angle $B < \text{angle } C$ and the values of B and C satisfy the equation $2 \tan x - k(1 + \tan^2 x) = 0$ where ($0 < k < 1$). Then the measure of angle A is :

त्रिभुज ABC में, कोण $B < \text{कोण } C$ यदि B तथा C समीकरण $2 \tan x - k(1 + \tan^2 x) = 0$ को सन्तुष्ट करता है जहाँ ($0 < k < 1$) तो कोण A का मान होगा :

(A) $\pi/3$ (B) $2\pi/3$ (C) $\pi/2$ (D) $3\pi/4$ **Ans.** C

Sol. $k = \frac{2 \tan x}{1 + \tan^2 x} = \sin 2x \Rightarrow \sin 2C = \sin 2B$

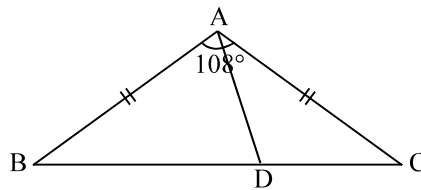
But $\angle C > \angle B$

$$2C = \pi - 2B \Rightarrow B + C = \pi/2$$

$$\therefore \angle A = \pi/2$$

10. In an isosceles triangle ABC, $AB = AC$, $\angle BAC = 108^\circ$ and AD trisects $\angle BAC$ and $BD > DC$. The ratio $\frac{BD}{DC}$ is:

यदि समद्विबाहु त्रिभुज ABC में, $AB = AC$, $\angle BAC = 108^\circ$ तथा AD $\angle BAC$ को समत्रिभाजित है तथा $BD > DC$ तो $\frac{BD}{DC}$ का मान होगा :



(A) $\frac{3}{2}$

(B) $\frac{\sqrt{5}+1}{2}$

(C) $\sqrt{5}-1$

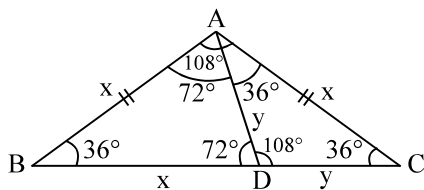
(D) 2

Ans. B

Sol. In $\triangle ABC$ ($\frac{x}{y} = ?$)

$$\frac{x+y}{\sin 108^\circ} = \frac{x}{\sin 36^\circ}$$

$$\frac{x+y}{x} = \frac{\sin 108^\circ}{\sin 36^\circ} \dots (1)$$



$$1 + \frac{y}{x} = \frac{\sin 72^\circ}{\sin 36^\circ} = 2 \cos 36^\circ$$

$$\frac{y}{x} = \frac{\sqrt{5}+1}{2} - 1 = \frac{\sqrt{5}-1}{2}$$

$$\frac{x}{y} = \frac{2(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)} = \frac{\sqrt{5}+1}{2}$$

11. The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. The ratio $A : B : C$ is :

त्रिभुज ABC की भुजाओं अनुपात $1 : \sqrt{3} : 2$ है तो अनुपात $A : B : C$ क्या होगा :

- (A) $3 : 5 : 2$ (B) $1 : \sqrt{3} : 2$ (C) $3 : 2 : 1$ (D) $1 : 2 : 3$

Ans. D

Sol. Let sides be $x, \sqrt{3}x, 2x$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = K$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{3x^2 + 4x^2 - x^2}{2(2x)(\sqrt{3}x)} = \frac{6x^2}{4\sqrt{3}x^2} = \frac{\sqrt{3}}{2}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{x^2 + 4x^2 - 3x^2}{2(x)(2x)} = \frac{2x^2}{4x^2} = \frac{1}{2}$$

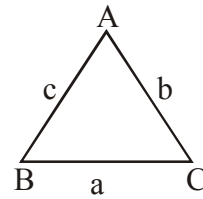
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{x^2 + 3x^2 - 4x^2}{2(x)(\sqrt{3}x)} = 0$$

$$\cos A : \cos B : \cos C = \frac{\sqrt{3}}{2} : \frac{1}{2} : 0$$

$$A : B : C = \frac{\pi}{6} : \frac{\pi}{3} : \frac{\pi}{2}$$

$$= \frac{1}{6} : \frac{1}{3} : \frac{1}{2}$$

$$= 1 : 2 : 3$$



12. In $\triangle ABC$, a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC. The correct relation is :

एक $\triangle ABC$ में a, b, c भुजाएँ एवं A, B, C कोण हैं। सही सम्बन्ध क्या होगा :

(A) $(b-c)\sin\left(\frac{B-C}{2}\right) = a \cos\left(\frac{A}{2}\right)$ (B) $(b-c)\cos\left(\frac{A}{2}\right) = a \sin\left(\frac{B-C}{2}\right)$

(C) $(b+c)\sin\left(\frac{B+C}{2}\right) = a \cos\left(\frac{A}{2}\right)$ (D) $(b-c)\cos\left(\frac{A}{2}\right) = 2a \sin\left(\frac{B+C}{2}\right)$

Ans. B

Sol. $(b-c)\sin\left(\frac{B-C}{2}\right) = 4 \cos \frac{A}{2} \Rightarrow 2R(\sin B - \sin C)\sin\left(\frac{B-C}{2}\right) = 2R \sin A \cos \frac{A}{2}$

$$\Rightarrow 2 \sin\left(\frac{B-C}{2}\right) \cos\left(\frac{B+C}{2}\right) \cdot \sin\left(\frac{B-C}{2}\right) = 2 \sin \frac{A}{2} \cos \frac{A}{2} \cos \frac{A}{2}$$

$$\Rightarrow \sin^2\left(\frac{B-C}{2}\right) = \cos^2\left(\frac{A}{2}\right) \Rightarrow \sin^2\left(\frac{B-C}{2}\right) = \sin^2\left(\frac{B+C}{2}\right) \Rightarrow \sin^2\left(\frac{B+C}{2}\right) - \sin^2\left(\frac{B-C}{2}\right) = 0$$

$$\Rightarrow \sin\left(\frac{B+C}{2} + \frac{B-C}{2}\right) \cdot \sin\left(\frac{B+C}{2} - \frac{B-C}{2}\right) = 0 \Rightarrow \sin B \cdot \sin C = 0 \text{ Not possible}$$

13. If in a ΔABC , $\cos A \cdot \cos B + \sin A \sin B \sin 2C = 1$ then, the statement which is incorrect, is :

- (A) ΔABC is isosceles but not right angled (B) ΔABC is acute angled
(C) ΔABC is right angled (D) least angle of the triangle is $\pi/4$

यदि ΔABC में $\cos A \cdot \cos B + \sin A \sin B \sin 2C = 1$ है तो कौनसा कथन असत्य है :

- (A) ΔABC समद्विबाहु परन्तु समकोण नहीं (B) ΔABC न्यून कोण त्रिभुज है
(C) ΔABC समकोण है (D) न्यूनतम कोण $\frac{\pi}{4}$ है

Ans. C

Sol. $\sin 2C = \frac{1 - \cos A \cos B}{\sin A \sin B} \leq 1$. Now proceed $A = B = \frac{3\pi}{8}$ and $C = \frac{\pi}{4}$

$$1 \leq \cos(A - B) \Rightarrow \cos(A - B) = 1 \Rightarrow A = B$$

14. The product of the arithmetic mean of the lengths of the sides of a triangle and harmonic mean of the lengths of the altitudes of the triangle is equal to :

त्रिभुज की भुजाओं के स.माध्य तथा शीर्ष लम्बों के ह.माध्य का गुणनफल निम्न में से किसके बराबर होगा :

- (A) Δ (B) 2Δ (C) 3Δ (D) 4Δ

Ans. B

$$\text{Sol.} : ah_1 = bh_2 = ch_3 = 2\Delta \Rightarrow \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} = \frac{a+b+c}{2\Delta} \Rightarrow \frac{a+b+c}{3} \cdot \frac{3}{\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}} = 2\Delta$$

15. A triangle has base 10 cm long and the base angles of 50° and 70° . If the perimeter of the triangle is $x + y \cos z^\circ$ where $z \in (0, 90)$ then the value of $x + y + z$ equals :

एक त्रिभुज का आधार 10 cm लम्बा तथा आधार कोण 50° तथा 70° है। त्रिभुज का परिमाप $x + y \cos z^\circ$ है जहाँ $z \in (0, 90)$ तो $x + y + z$ का मान होगा :

- (A) 60 (B) 55 (C) 50 (D) 40

Ans. D

Sol. Given $P = x + y \cos z^\circ$
 Also $P = 10 + b + c$ (1)

Now using sine law

$$\frac{10}{\sin 60^\circ} = \frac{b}{\sin 50^\circ} = \frac{c}{\sin 70^\circ}$$

$$\therefore b = \frac{10 \sin 50^\circ}{\sin 60^\circ} \quad \text{and} \quad c = \frac{10 \sin 70^\circ}{\sin 60^\circ}$$

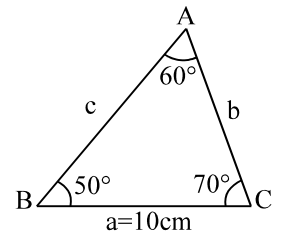
$$b + c = \frac{10}{\sin 60^\circ} (\sin 50^\circ + \sin 70^\circ)$$

$$= \frac{10}{\sin 60^\circ} [2 \sin 60^\circ \cdot \cos 10^\circ] = 20 \cos 10^\circ$$

$$\therefore P = 10 + 20 \cos 10^\circ$$

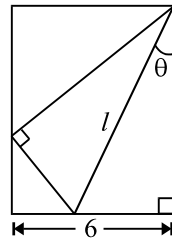
$$\therefore x = 10; \quad y = 20; \quad z = 10$$

$$x + y + z = 40$$



16. One side of a rectangular piece of paper is 6 cm, the adjacent sides being longer than 6 cms. One corner of the paper is folded so that it sets on the opposite longer side. If the length of the crease is l cms and it makes an angle θ with the long side as shown, then l is :

कागज के एक आयताकार टुकड़े की एक भुजा 6 cms है तथा दूसरी भुजा 6 cm से अधिक है। कागज के एक कोने को मोड़कर विपरीत भुजा के एक बिन्दु पर रखा जाता है। यदि शिकन (crease) की लम्बाई l cms है तथा यह दीर्घ भुजा के साथ कोण θ बनाता है। तो l होगा :



(A) $\frac{3}{\sin \theta \cos^2 \theta}$ (B) $\frac{6}{\sin^2 \theta \cos \theta}$ (C) $\frac{3}{\sin \theta \cos \theta}$ (D) $\frac{3}{\sin^3 \theta}$

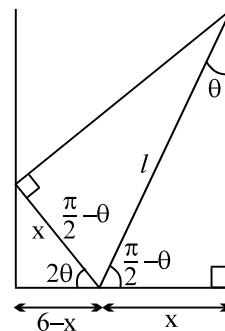
Ans. A

Sol. $\sin \theta = \frac{x}{l}$ (1); also $\cos 2\theta = \frac{6-x}{x}$

$$1 + \cos 2\theta = \frac{6}{x};$$

$$2 \cos^2 \theta = \frac{6}{l \sin \theta} \quad \{\text{substituting } x = l \sin \theta \text{ from (1)}\}$$

$$l = \frac{3}{\sin \theta \cos^2 \theta} \text{ Ans.]}$$



17. Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \sqrt{\frac{3}{2}}$, then $\angle A$ is equal to :

त्रिभुज ABC के कोण A, B तथा C समान्तर श्रेणी में हैं। यदि $\frac{b}{c} = \sqrt{\frac{3}{2}}$ है, तो $\angle A$ बराबर है :

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. A, B, C $\rightarrow$ AP $\Rightarrow B = 60^\circ$; $\frac{b}{c} = \sqrt{\frac{3}{2}}$

$$\Rightarrow \frac{\sin B}{b} = \frac{\sin C}{c} \Rightarrow \sin C = \frac{\sin B}{\left(\frac{b}{c}\right)} = \frac{\sqrt{3}}{2} \times \sqrt{\frac{2}{3}} = \frac{1}{\sqrt{2}} \Rightarrow C = 45^\circ$$

$$\Rightarrow A = 180 - 105 \Rightarrow A = 75^\circ = \frac{5\pi}{12} \text{ (C)}$$

18. In a triangle ABC, $\angle A = 60^\circ$ and $b : c = (\sqrt{3} + 1) : 2$ then $(\angle B - \angle C)$ has the value equal to :

एक त्रिभुज ABC में $\angle A = 60^\circ$ तथा $b : c = (\sqrt{3} + 1) : 2$ है, तो $(\angle B - \angle C)$ का मान है :

- (A) 15° (B) 30° (C) 22.5° (D) 45°

Ans. B

Sol. $A = 60^\circ \Rightarrow B + C = 120^\circ$; $B - C = ?$

$$\frac{b}{c} = \frac{\sqrt{3} + 1}{2} = \frac{\sin B}{\sin C}, \frac{a}{b} = \frac{\sqrt{3}}{2 \sin B} \Rightarrow \sin B = \frac{b\sqrt{3}}{2a}$$

$$\Rightarrow \frac{a}{c} = \frac{\sqrt{3}}{2 \sin C} \Rightarrow \sin C = \frac{c\sqrt{3}}{2a}$$

$$\text{Tangent Rule: } \tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right) \cot\left(\frac{A}{2}\right)$$

$$\Rightarrow \tan\left(\frac{B-C}{2}\right) \cdot \frac{1}{\sqrt{3}} = \frac{\sqrt{3}-1}{\sqrt{3}+3} = \frac{(\sqrt{3}-1)}{\sqrt{3}(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} = \frac{4-2\sqrt{3}}{2 \times \sqrt{3}} = \frac{2-\sqrt{3}}{\sqrt{3}}$$

$$\Rightarrow \tan\left(\frac{B-C}{2}\right) = 2-\sqrt{3} = \tan(15^\circ) \Rightarrow B-C = 30^\circ \text{ (B)}$$

19. In a triangle ABC, if $A = 30^\circ$ and $\frac{b}{c} = \frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$, then C, is equal to :

एक त्रिभुज ABC में, यदि $A = 30^\circ$ तथा $\frac{b}{c} = \frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$ है, तो C का मान है :

- (A) 75° (B) 15° (C) $52\frac{1}{2}^\circ$ (D) $97\frac{1}{2}^\circ$

Ans. C

Sol. $A = 30^\circ$; $\frac{b}{c} = \frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$

$$\tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right) \cot \frac{A}{2} = \left(\frac{2\sqrt{2}-2}{4+2\sqrt{3}}\right) \times \left(\frac{1}{2-\sqrt{3}}\right) = \frac{\sqrt{2}-1}{1}$$

$$\Rightarrow \frac{B-C}{2} = \frac{45}{2}$$

$$\Rightarrow B-C = 45$$

$$\Rightarrow B+C = 150$$

$$2C = 105 \Rightarrow C = 52.5^\circ \text{ (C)}$$

20. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides

opposite to A, B and C respectively, then the value of the expression $\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$, is :

यदि त्रिभुज में कोण A, B तथा C समान्तर श्रेणी में है तथा यदि a, b तथा c उनके सामने वाली भुजायें हैं तो व्यंजक $\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$

का मान है :

- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

Ans. D

Sol. $B = 60^\circ$

$$\therefore \frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A = 2 \sin A \cos C + 2 \sin C \cos A$$

$$= 2 \sin(A + C) = 2 \sin B = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

21. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are :

माना ABC एक त्रिभुज है जिसके लिये $\angle ACB = \frac{\pi}{6}$ है तथा a, b तथा c शीर्षों A, B तथा C के सामने वाली भुजाओं की लम्बाई है।

तो x के मान जिनके लिये $a = x^2 + x + 1$, $b = x^2 - 1$ तथा $c = 2x + 1$ है :

- (A) $-(2 + \sqrt{3})$ (B) $1 + \sqrt{3}$ (C) $2 + \sqrt{3}$ (D) $4\sqrt{3}$

Ans. B

Sol. $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

$$\frac{\sqrt{3}}{2} = \frac{(x^2 + x + 1)^2 + (x^2 - 1)^2 - (2x + 1)^2}{2(x^2 + x + 1)(x^2 - 1)}$$

$$\Rightarrow x^4(2 - \sqrt{3}) + x^3(2 - \sqrt{3}) - 3x^2 - x(2 - \sqrt{3}) + (\sqrt{3} + 1) = 0$$

$$\Rightarrow (x^2 - 1)[(2 - \sqrt{3})x^2 + (2 - \sqrt{3})x - (\sqrt{3} + 1)] = 0$$

Now $\begin{cases} x^2 + x + 1 + x^2 - 1 > 2x + 1 \\ x^2 + x + 1 + 2x + 1 > x^2 - 1 \\ 2x + 1 + x^2 - 1 > x^2 + x + 1 \end{cases} \quad (\because \text{sum of two sides is greater than third side})$

$$\Rightarrow x > 1 \quad \Rightarrow x = 1 + \sqrt{3}$$

Alternate :

$$\tan \frac{\pi}{12} = \sqrt{\frac{(s-b)(s-a)}{s(s-c)}}$$

$$s = x^2 + \frac{3x+1}{2}, s-b = \frac{3(x+1)}{2}, s-a = \frac{x-1}{2}, s-c = x^2 - \frac{(x+1)}{2}$$

$$\tan \frac{\pi}{12} = \sqrt{\frac{\frac{3}{2}(x+1)\frac{(x-1)}{2}}{\left(x^2 + \frac{3x+1}{2}\right)\left(x^2 - \frac{x+1}{2}\right)}}$$

Simplifying

$$2 - \sqrt{3} = \frac{\sqrt{3}}{2x+1}$$

$$x = \sqrt{3} + 1$$

HALF ANGLE FORMULAE

22. The expression $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to :

व्यंजक $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ का मान होगा :

- (A) $\cos 2A$ (B) $1 + \cos A$ (C) $1 - \cos A$ (D) $\sin^2 A$

Ans. D

Sol.
$$\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$$

$$\frac{2s(2s-2a)(2s-2b)(2s-2c)}{4b^2c^2}$$

$$\frac{16s(s-a)(s-b)(s-c)}{4b^2c^2} \Rightarrow \frac{4\Delta^2}{b^2c^2}$$

$$\Rightarrow \frac{4\left(\frac{1}{2}bc \sin A\right)^2}{b^2c^2} \Rightarrow \sin^2 A$$

23. In a ΔABC , if $\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13}$, then $\tan^2 \frac{A}{2}$ is equal to :

यदि ΔABC , में $\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13}$, हो तो $\tan^2 \frac{A}{2}$ का मान होगा :

- (A) $\frac{143}{342}$ (B) $\frac{13}{33}$ (C) $\frac{11}{39}$ (D) $\frac{12}{37}$

Ans. B

Sol.
$$\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13} = k$$

$$s = 11k + a \quad s = 12k + b \quad s = 13k + c$$

$$3s - 2s = (11 + 12 + 13)k$$

$$s = 36k$$

$$\tan^2 \frac{A}{2} = \frac{(s-b)(s-c)}{s(s-a)}$$

$$= \frac{12k \times 13k}{36k \times 11k} = \frac{13}{33}$$

24. If a, b, c are in A.P. then the numerical value of $\tan \frac{A}{2} \tan \frac{C}{2}$ is:

यदि a, b, c स.श्रे. में हो तो $\tan \frac{A}{2} \tan \frac{C}{2}$ का मान होगा :

- (A) 1/4 (B) 1/3 (C) 1/2 (D) 1

Ans. B

Sol. $\tan \frac{A}{2} \tan \frac{C}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \times \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

$$= \frac{s-b}{s} \Rightarrow \frac{2s-2b}{2s} = \frac{a+b+c-a-c}{a+b+c} = \frac{b}{3b} = \frac{1}{3}$$

25. In ΔABC , the minimum value of $\frac{\sum \cot^2 \frac{A}{2} \cdot \cot^2 \frac{B}{2}}{\prod \cot^2 \frac{A}{2}}$ is:

- (A) 1 (B) 2 (C) 3 (D) non existent

यदि ΔABC है तो $\frac{\sum \cot^2 \frac{A}{2} \cdot \cot^2 \frac{B}{2}}{\prod \cot^2 \frac{A}{2}}$ का न्यूनतम मान होगा :

- (A) 1 (B) 2 (C) 3 (D) विद्यमान नहीं है

Ans. A

Sol. $\frac{\sum \cot^2 \frac{A}{2} \cot^2 \frac{B}{2}}{\prod \cot^2 \frac{A}{2}}$

$$\Rightarrow \left[\frac{1}{\tan^2 \frac{A}{2} \tan^2 \frac{B}{2}} + \frac{1}{\tan^2 \frac{B}{2} \tan^2 \frac{C}{2}} + \frac{1}{\tan^2 \frac{C}{2} \tan^2 \frac{A}{2}} \right] \times \pi \tan^2 \frac{A}{2}$$

$$\Rightarrow \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2}$$

$$\Rightarrow \left(\tan \frac{A}{2} - \tan \frac{B}{2} \right)^2 \geq 0$$

$$\Rightarrow \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} \geq 2 \tan \frac{A}{2} \tan \frac{B}{2}$$

$$\Rightarrow 2 \sum \tan^2 \frac{A}{2} \geq 2 \sum \tan \frac{A}{2} \tan \frac{B}{2}$$

In $\triangle ABC$, so sum of $\tan \frac{A}{2} \tan \frac{B}{2}$ is always greater than 1

$$\sum \tan^2 \frac{A}{2} \geq 1$$

min. value = 1

BASED ON CIRCLES ASSOCIATED WITH A TRIANGLE

26. In a triangle ABC, right angles at B, the inradius is :

यदि त्रिभुज ABC में B समकोण है तो अन्तःवृत्त की त्रिज्या होगी :

(A) $\frac{AB+BC-AC}{2}$ (B) $\frac{AB+AC-BC}{2}$ (C) $\frac{AB+BC+AC}{2}$ (D) $\frac{BC+AC-AB}{2}$

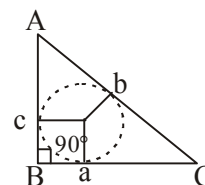
Ans. A

Sol. $b^2 = a^2 + c^2$

$$r = (s - b) \tan \frac{B}{2}$$

$$r = \frac{2s - 2b}{2} = \frac{a + c - b}{2}$$

$$r = \frac{AB + BC - AC}{2}$$



27. In a $\triangle ABC$, $\frac{r_1 + r_2}{1 + \cos C}$ is equal to :

त्रिभुज $\triangle ABC$ में, $\frac{r_1 + r_2}{1 + \cos C}$ का मान होगा :

(A) $\frac{abc}{2\Delta}$ (B) $2R$ (C) $\frac{abc}{\Delta^2}$ (D) $\frac{abc}{c\Delta}$

Ans. A

Sol.
$$\frac{r_1 + r_2}{1 + \cos c} = \frac{r_1 + r_2}{2 \cos^2 \frac{c}{2}} = \frac{\frac{\Delta}{s-a} + \frac{\Delta}{s-b}}{2 \cos^2 \frac{c}{2}}$$

$$\Rightarrow \frac{\Delta}{2} \left(\frac{2s-a-b}{(s-a)(s-b)} \right) \times \frac{ab}{s(s-c)}$$

$$\Rightarrow \frac{\Delta}{2} \cdot \frac{abc}{\Delta^2} \Rightarrow \frac{abc}{2\Delta}$$

28. In a $\triangle ABC$, $r_1 r_2 + r r_3 =$

त्रिभुज $\triangle ABC$ में, $r_1 r_2 + r r_3 =$

- (A) $\frac{ab}{c}$ (B) abc (C) ab (D) $\frac{(a+b)}{c}$

Ans. C

Sol. $r_1 r_2 + r r_3$

$$\frac{\Delta}{s-a} \times \frac{\Delta}{s-b} + \frac{\Delta}{s} \times \frac{\Delta}{(s-c)}$$

$$\Delta^2 \left(\frac{1}{(s-a)(s-b)} + \frac{1}{s(s-c)} \right)$$

$$\Delta^2 \left(\frac{s(s-c) + (s-a)(s-b)}{s(s-a)(s-b)(s-c)} \right)$$

$$s^2 - sc + s^2 - sb - sa + ab$$

$$2s^2 - s(a+b+c) + ab$$

$$2s^2 - 2s^2 + ab = ab$$

29. In a triangle ABC, $R(b+c) = a\sqrt{bc}$ where R is the circumradius of the triangle. Then the triangle is :

- (A) Isosceles but not right (B) right but not isosceles
(C) right isosceles (D) equilateral

त्रिभुज ABC में यदि $R(b+c) = a\sqrt{bc}$ जहाँ R परिवृत्त की त्रिज्या है तो त्रिभुज होगा :

- (A) समद्विबाहु परन्तु समकोण नहीं (B) समकोण लेकिन समद्विबाहु नहीं
(C) समकोण तथा समद्विबाहु (D) समबाहु

Ans. C

Sol. $R(b+c) = a\sqrt{bc}$

$$R(b+c) = 2R \sin A \sqrt{bc}$$

$$\therefore \sin A = \frac{b+c}{2\sqrt{bc}}$$

now applying $AM \geq GM$ for b and c

$$\frac{b+c}{2} \geq \sqrt{bc} \Rightarrow b+c \geq 2\sqrt{bc}$$

hence $\sin A \geq 1$ which is not possible.

hence $\sin A = 1 \Rightarrow A = 90^\circ$

$\therefore A = 90^\circ$ and $b = c \Rightarrow (C)$

30. A triangle has sides 6, 7, 8. The line through its incentre parallel to the shortest side is drawn to meet the other two sides at P and Q. The length of the segment PQ is :

एक त्रिभुज की भुजाएँ 6, 7, 8 हैं इसके अन्तः केन्द्र से एक रेखा त्रिभुज की सबसे छोटी भुजा के समान्तर खींची गई है जो अन्य दो भुजाओं को P तथा Q पर काटती है तो रेखाखण्ड PQ का मान होगा :

- (A) $\frac{12}{5}$ (B) $\frac{15}{4}$ (C) $\frac{30}{7}$ (D) $\frac{33}{9}$

Ans. C

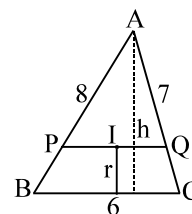
Sol. $\Delta = \frac{6 \cdot h}{2} = 3h$; (where h is the altitude from A)

Also $\Delta = \frac{21 \cdot r}{2}$ (using $\Delta = r \cdot s$)

$$\therefore \frac{21 \cdot r}{2} = \frac{6 \cdot h}{2} = 3h \Rightarrow \frac{r}{h} = \frac{2}{7}$$

now APQ and ABC are similar

$$\frac{h-r}{h} = \frac{PQ}{6} \Rightarrow 1 - \frac{r}{h} = \frac{PQ}{6} \Rightarrow 1 - \frac{2}{7} = \frac{PQ}{6} \Rightarrow \frac{5}{7} = \frac{PQ}{6} \Rightarrow PQ = \frac{30}{7}$$



31. If r_1, r_2, r_3 be the radii of excircles of the triangle ABC, then $\frac{\sum r_i}{\sqrt{\sum r_i r_j}}$ is equal to :

यदि r_1, r_2, r_3 त्रिभुज ABC के बाह्य वृत्तों की त्रिज्या हो तो $\frac{\sum r_i}{\sqrt{\sum r_i r_j}}$ का मान होगा :

(A) $\sum \cot \frac{A}{2}$ (B) $\sum \cot \frac{A}{2} \cot \frac{B}{2}$ (C) $\sum \tan \frac{A}{2}$ (D) $\prod \tan \frac{A}{2}$

Ans. C

Sol. $\frac{s \sum \tan \frac{A}{2}}{\sqrt{s^2}} = \sum \tan \frac{A}{2} \Rightarrow C$

32. If x, y & z are the distances of incentre from the vertices of the ΔABC respectively then $\frac{abc}{xyz} =$

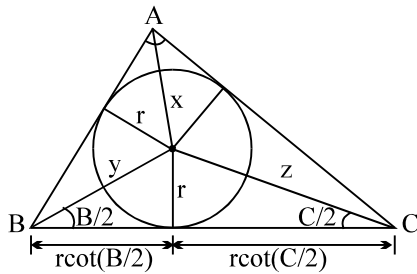
यदि x, y तथा z त्रिभुज ABC में अन्तः केन्द्र की शीर्षों से दूरियां हैं तो $\frac{abc}{xyz}$ का मान होगा :

(A) $\prod \tan \frac{A}{2}$ (B) $\sum \cot \frac{A}{2}$ (C) $\sum \tan \frac{A}{2}$ (D) $\sum \sin \frac{A}{2}$

Ans. B

Sol. $x = r \operatorname{cosec} \frac{A}{2}$

$$a = r \left(\cot \frac{B}{2} + \cot \frac{C}{2} \right)$$



$$\frac{a}{x} = \left(\cot \frac{B}{2} + \cot \frac{C}{2} \right) \cdot \sin \frac{A}{2} = \frac{\sin \frac{A}{2} \cdot \cos \frac{A}{2}}{\sin \frac{B}{2} \cdot \sin \frac{C}{2}}$$

$$\therefore \frac{abc}{xyz} = \frac{\cos \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2}}{\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}} = \cot \frac{A}{2} \cdot \cot \frac{B}{2} \cdot \cot \frac{C}{2}$$

In a triangle $\prod \cot \frac{A}{2} = \sum \cot \frac{A}{2}$

33. With usual notations, in a triangle ABC, $a \cos(B - C) + b \cos(C - A) + c \cos(A - B)$ is equal to :

सामान्य त्रिभुज के प्रतिकों के अनुसार त्रिभुज ABC में $a \cos(B - C) + b \cos(C - A) + c \cos(A - B)$ का मान होगा :

- (A) $\frac{abc}{R^2}$ (B) $\frac{abc}{4R^2}$ (C) $\frac{4abc}{R^2}$ (D) $\frac{abc}{2R^2}$

Ans. A

Sol. Put $a = 2R \sin A$ etc.

$$E = R[\sin 2B + \sin 2C + \sin 2C + \sin 2A + \sin 2A + \sin 2B] \\ = 2R(\sin 2A + \sin 2B + \sin 2C)$$

$$= 8R \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} = \frac{abc}{R^2} \text{ Ans.}$$

Alternatively: $a = 2R \sin A = 2R \sin(B + C)$

$$R[(2 \sin(B + C) \cos(B - C) + \dots + \dots)]$$

$$R[(\sin 2B + \sin 2C) + \dots + \dots]$$

$$2R(\sin 2A + \sin 2B + \sin 2C)$$

$$8R \sin A \cdot \sin B \cdot \sin C = \frac{abc}{R^2}$$

34. Given an isosceles triangle, whose one angle is 120° and radius of its incircle is $\sqrt{3}$. Then the area of triangle in sq. units is :

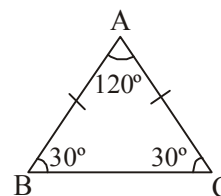
एक समद्विबाहु त्रिभुज में एक कोण 120° है तथा अन्तः वृत्त की त्रिज्या $\sqrt{3}$ है। तो उसका क्षेत्रफल है :

- (A) $7 + 12\sqrt{3}$ (B) $12 - 7\sqrt{3}$ (C) $12 + 7\sqrt{3}$ (D) 4π

Ans. C

Sol. $\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ}$

$$\frac{a}{\sqrt{3}} = \frac{b}{1} = \frac{c}{1}$$



$$a : b : c = \sqrt{3} : 1 : 1$$

$$a = \sqrt{3}k$$

$$b = k$$

$$c = k$$

$$r = (s - a) \tan \frac{A}{2}$$

$$\sqrt{3} = \left(\frac{\sqrt{3}k + k + k}{2} - \sqrt{3}k \right) \tan \frac{120}{2}$$

$$\frac{\sqrt{3}k + 2k}{2} = \sqrt{3}k + 1$$

$$\sqrt{3}k + 2k = 2\sqrt{3}k + 2$$

$$-\sqrt{3}k + 2k = 2$$

$$k = \frac{2}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} \Rightarrow \frac{2(2+\sqrt{3})}{4-3} \Rightarrow 2(2+\sqrt{3})$$

$$r = \frac{\Delta}{s} \Rightarrow \Delta = sr \Rightarrow \Delta = \frac{(\sqrt{3}+2)k}{2} \times \sqrt{3}$$

$$= \frac{2(\sqrt{3}+2)(2+\sqrt{3})}{2} \times \sqrt{3}$$

$$= (4+3+4\sqrt{3}) \times \sqrt{3}$$

$$= (7+4\sqrt{3}) \times \sqrt{3} \Rightarrow 12+7\sqrt{3}$$

LENGTH OF ANGLE BISECTORS AND MEDIAN

35. Triangle ABC is right angled at A. The points P and Q are on the hypotenuse BC such that BP = PQ = QC. If AP = 3 and AQ = 4 then the length BC is equal to :

त्रिभुज ABC में A कोण समकोण है। बिन्दु P तथा Q, कर्ण BC पर इस प्रकार हैं BP = PQ = QC कि यदि AP = 3 तथा AQ = 4 तो BC की लम्बाई होगी :

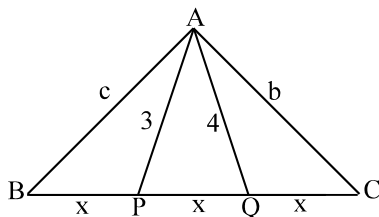
- (A) $\sqrt{27}$ (B) $\sqrt{36}$ (C) $\sqrt{45}$ (D) $\sqrt{54}$

Ans. C

Sol. In $\triangle ABP$

$$9 = c^2 + x^2 - 2cx \cos B; \text{ but } \cos B = \frac{c}{3x}$$

$$9 = c^2 + x^2 - 2cx \frac{c}{3x}$$



$$9 = x^2 + \frac{c^2}{3} \quad \dots(1) ;$$

$$\parallelly \quad 16 = x^2 + \frac{b^2}{3} \quad \dots(2)$$

$$(1) + (2)$$

$$25 = 2x^2 + \frac{1}{3}(b^2 + c^2) = 2x^2 + 3x^2$$

$$\Rightarrow 5x^2 = 25 \quad \Rightarrow x = \sqrt{5} ; \quad \therefore BC = 3\sqrt{5} = \sqrt{45}$$

ORTHOCENTER AND PEDAL TRIANGLE

36. In an acute angled triangle ABC, point D, E and F are the feet of the perpendiculars from A, B and C onto BC, AC and AB respectively. H is the intersection of AD and BE. If $\sin A = 3/5$ and $BC = 39$, the length of AH is :

किसी न्यून कोण त्रिभुज ABC में बिन्दु D, E तथा F शीर्षों A, B तथा C से भुजाओं BC, AC तथा AB पर लम्ब पाद हैं। H, AD तथा BE का प्रतिच्छेदन बिन्दु है यदि $\sin A = 3/5$ तथा $BC = 39$, तो AH की लम्बाई होगी :

- (A) 45 (B) 48 (C) 52 (D) 54

Ans. C

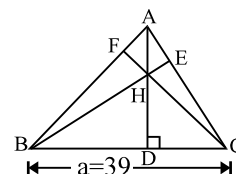
Sol. $AH = 2R \cos A$ (distance of orthocentre from the vertex)

given $\sin A = 3/5$; $\cos A = 4/5$

$$a = 39$$

$$\therefore \frac{a}{\sin A} = 2R; \quad \frac{39 \cdot 5}{3} = 2R \quad \Rightarrow 2R = 65$$

$$\therefore AH = 65 \cdot \frac{4}{5} = 52$$



REGULAR POLYGON

37. If R is the radius of circumscribing circle of a regular polygon of n-sides with side length 'a' then R = यदि R, n भुजाओं वाले समबहुभुज के परिवृत्त की त्रिज्या हो तो R बराबर होगा :

- (A) $\frac{a}{2} \sin \left(\frac{\pi}{n} \right)$ (B) $\frac{a}{2} \sin \left(\frac{\pi}{n} \right)$ (C) $\frac{a}{2} \operatorname{cosec} \left(\frac{\pi}{n} \right)$ (D) $\frac{a}{2} \operatorname{cosec} \left(\frac{\pi}{2n} \right)$

Ans. C

$$\text{Sol. } \sin \theta = \frac{P}{H} \quad 2\theta = \frac{2\pi}{n}$$

$$\sin \frac{\pi}{n} = \frac{\frac{a}{2}}{R}$$

$$R = \frac{a}{2} \operatorname{cosec} \frac{\pi}{n}$$

38. If A_1, A_2, A_3 denote respectively the areas of an inscribed polygon of $2n$ sides, inscribed polygon of n sides and circumscribed polygon of n sides, then A_2, A_1, A_3 are in :

(A) AP (B) GP (C) HP (D) none of these

यदि A_1, A_2, A_3 क्रमशः एक वृत्त में $2n$ भुजाओं वाले अन्तःसमबहुभुज (inscribed polygon), n भुजाओं वाले अन्तः समबहुभुज तथा n भुजाओं वाले परिगत समबहुभुज के क्षेत्रफलों को दर्शाता है तो A_2, A_1, A_3 किस में होंगे :

(A) AP (B) GP (C) HP (D) इनमें से कोई नहीं

Ans. B

Sol. $A_1 = \frac{2nR^2}{2} \sin \frac{\pi}{n}$

$$A_2 = \frac{nR^2}{2} \sin \left(\frac{2\pi}{n} \right)$$

$$A_3 = nr^2 \tan \frac{\pi}{n} = nR^2 \tan \frac{\pi}{n}$$

$$\frac{2nR^2}{2} \sin \frac{\pi}{n}, \frac{nR^2}{2} \sin \frac{2\pi}{n}, nR^2 \tan \frac{\pi}{n}$$

$$\frac{2}{2} \sin \frac{\pi}{n}, \frac{2}{2} \sin \frac{\pi}{n} \cos \frac{\pi}{n}, \frac{\sin \frac{\pi}{n}}{\cos \frac{\pi}{n}}$$

$$\cos \frac{\pi}{n}, 1, \frac{1}{\cos \frac{\pi}{n}} \rightarrow \text{G.P.}$$

39. The area of the circle and the area of a regular polygon of n sides and of perimeter equal to that of the circle are in the ratio of:

एक n भुजाओं वाले समबहुभुज का परिमाण वृत्त के परिमाण के समान है, तो वृत्त तथा समबहुभुज के क्षेत्रफल का अनुपात होगा :

(A) $\tan \left(\frac{\pi}{n} \right) : \frac{\pi}{n}$ (B) $\cos \left(\frac{\pi}{n} \right) : \frac{\pi}{n}$ (C) $\sin \frac{\pi}{n} : \frac{\pi}{n}$ (D) $\cot \left(\frac{\pi}{n} \right) : \frac{\pi}{n}$

Ans. A

Sol. $2\pi r = na$

$$A_c = \pi r^2 \Rightarrow \pi \left(\frac{na}{2\pi} \right)^2$$

$$A_p = \frac{na^2}{4} \cot \frac{\pi}{n}$$

$$\frac{A_c}{A_p} = \frac{\pi \left(\frac{na}{2\pi} \right)^2}{\frac{na^2}{4} \cot \frac{\pi}{n}} \Rightarrow \frac{\pi \times \frac{n^2 a^2}{4\pi^2}}{\frac{na^2}{4} \cot \frac{\pi}{n}} \Rightarrow \frac{n}{\pi \cot \frac{\pi}{n}} \Rightarrow \frac{\tan \frac{\pi}{n}}{\frac{\pi}{n}}$$